Amazon S3 Assignment 1

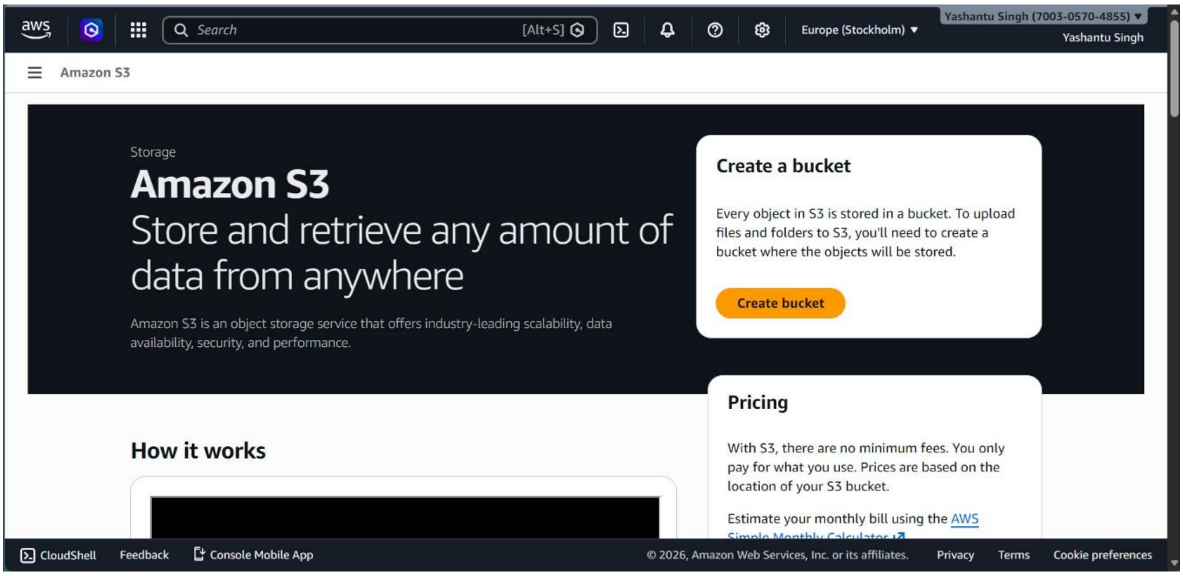
How to Create and Configure an Amazon S3 Bucket

Introduction to Amazon S3 and buckets

Amazon Simple Storage Service (Amazon S3) is a scalable object storage service that allows users to store and retrieve any amount of data from anywhere. An S3 bucket is a container for storing objects (files), and each bucket is uniquely named within AWS. Buckets are used to organize and manage data in the cloud.

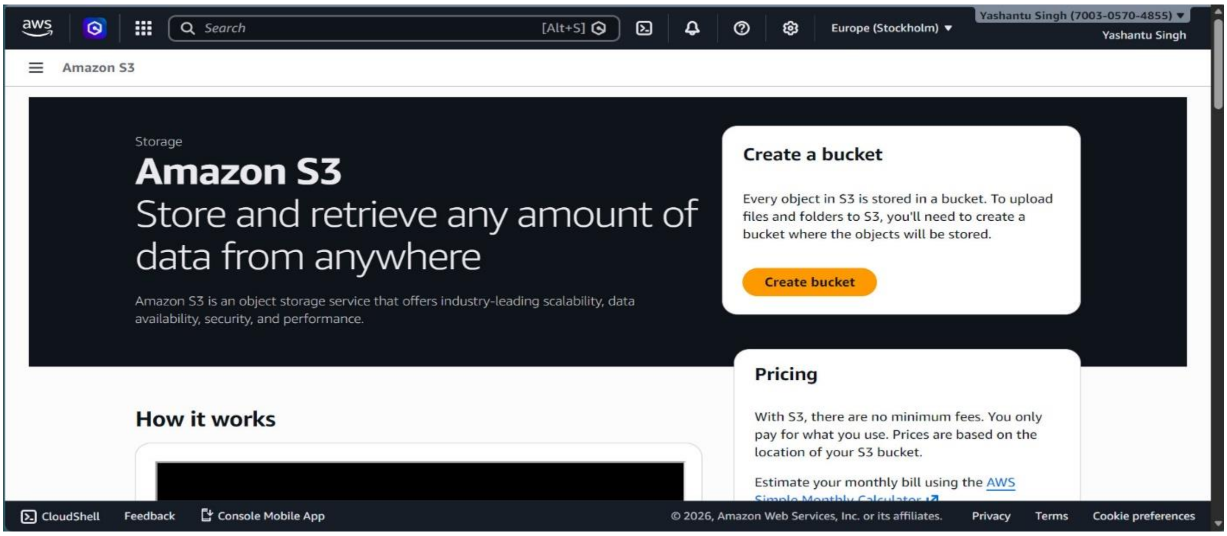
Step 1: Navigate to Amazon S3 service in AWS Console

Log in to your AWS Management Console. From the AWS services menu, search for and select 'S3' to open the Amazon S3 dashboard.



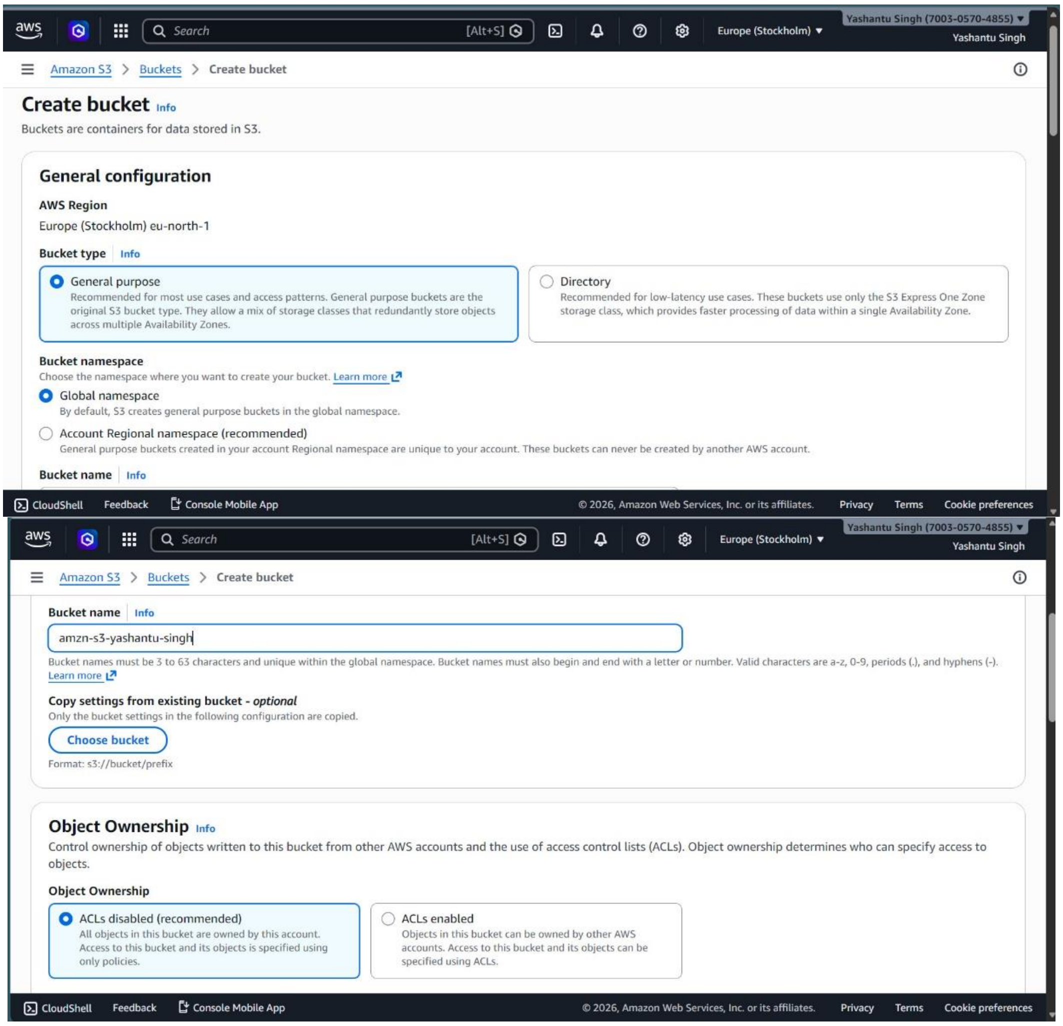
Step 2: Click 'Create bucket'

On the Amazon S3 dashboard, click the orange 'Create bucket' button to start the process of creating a new storage bucket.



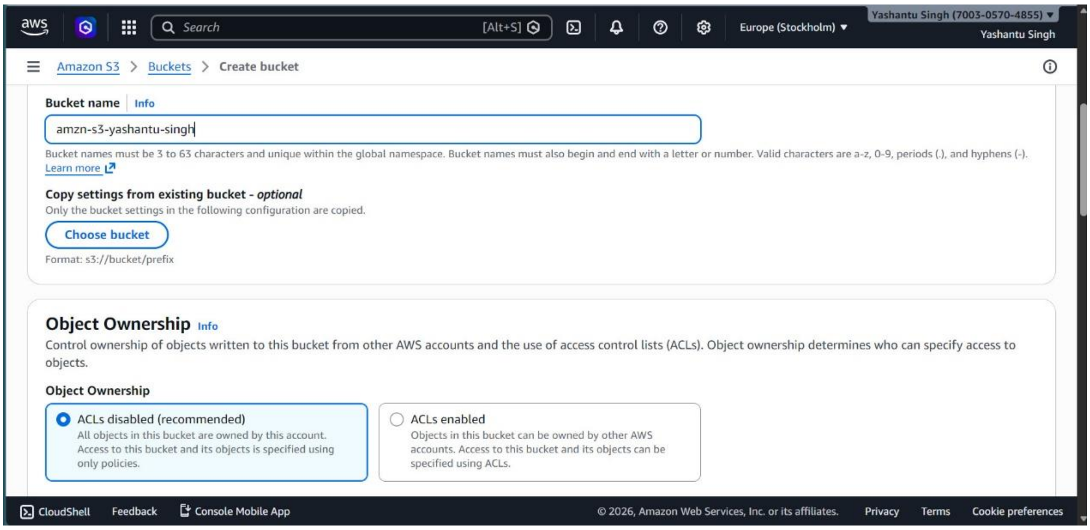
Step 3: Configure general settings (Region, Bucket type, Namespace, Bucket name)

Choose the AWS Region where your bucket will reside. Select the bucket type (General purpose or Directory). Choose the namespace (Global or Account Regional). Enter a unique bucket name following AWS naming conventions.



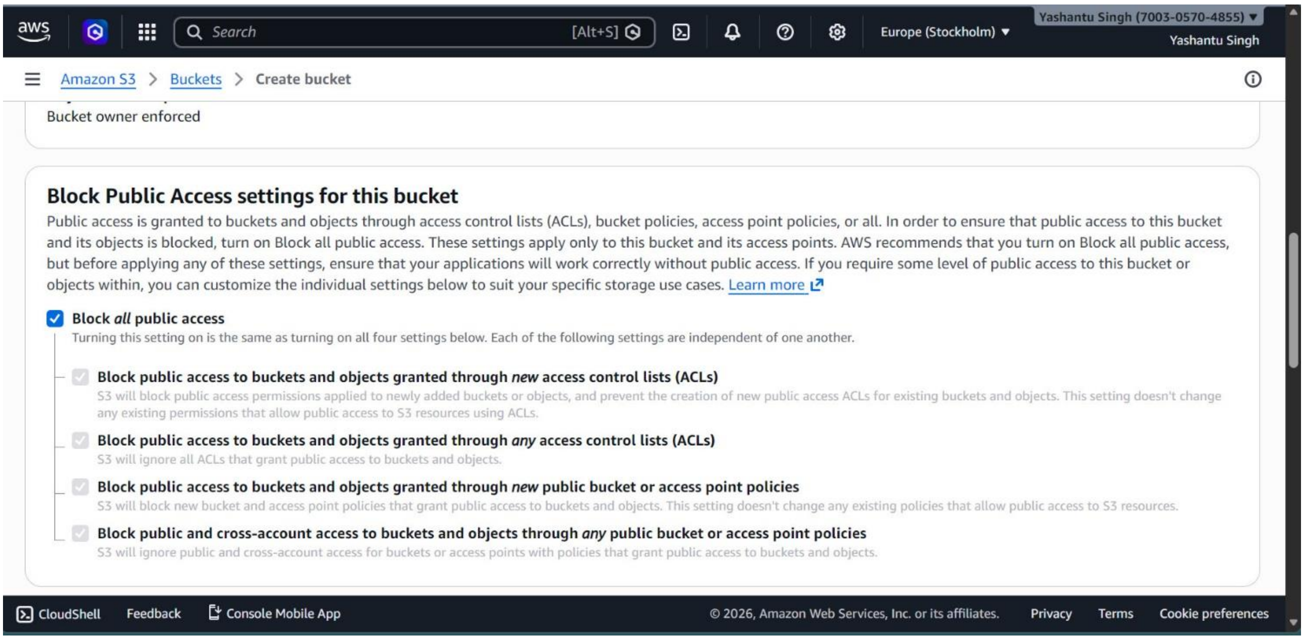
Step 4: Set object ownership (ACLs disabled recommended)

Choose the object ownership setting. It is recommended to disable ACLs to ensure all objects are owned by the bucket owner and access is managed through policies.



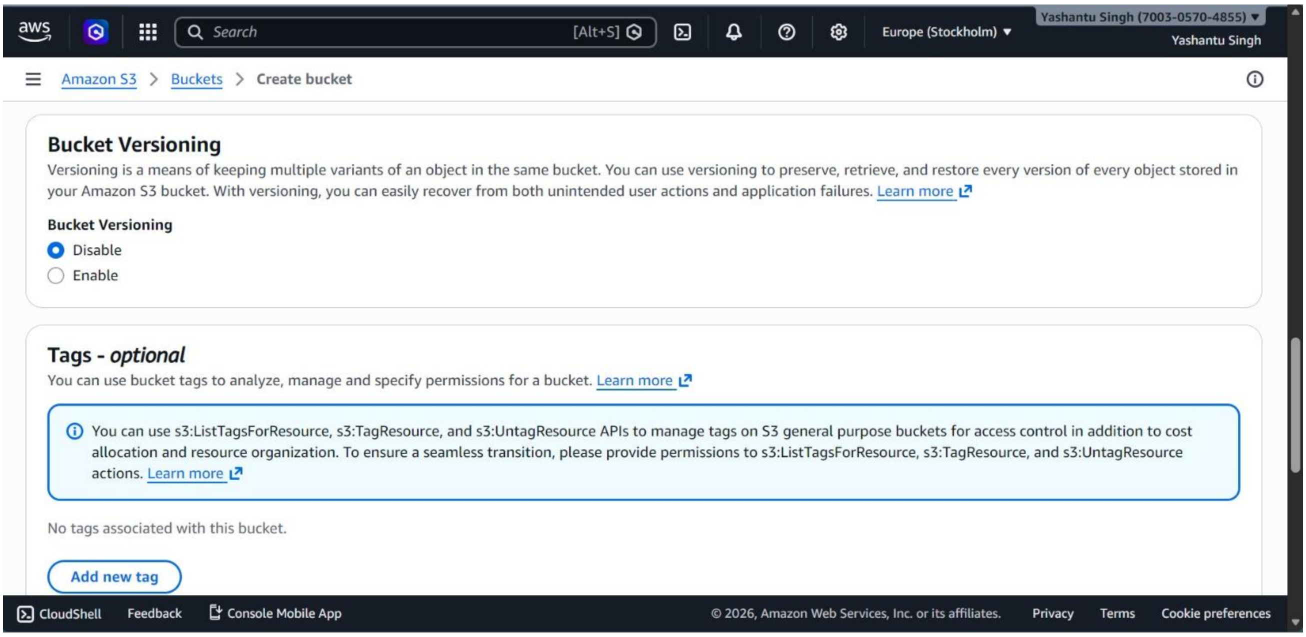
Step 5: Configure public access settings (Block all public access recommended)

Enable 'Block all public access' to prevent unauthorized access to your data. This is a recommended best practice for securing your data.



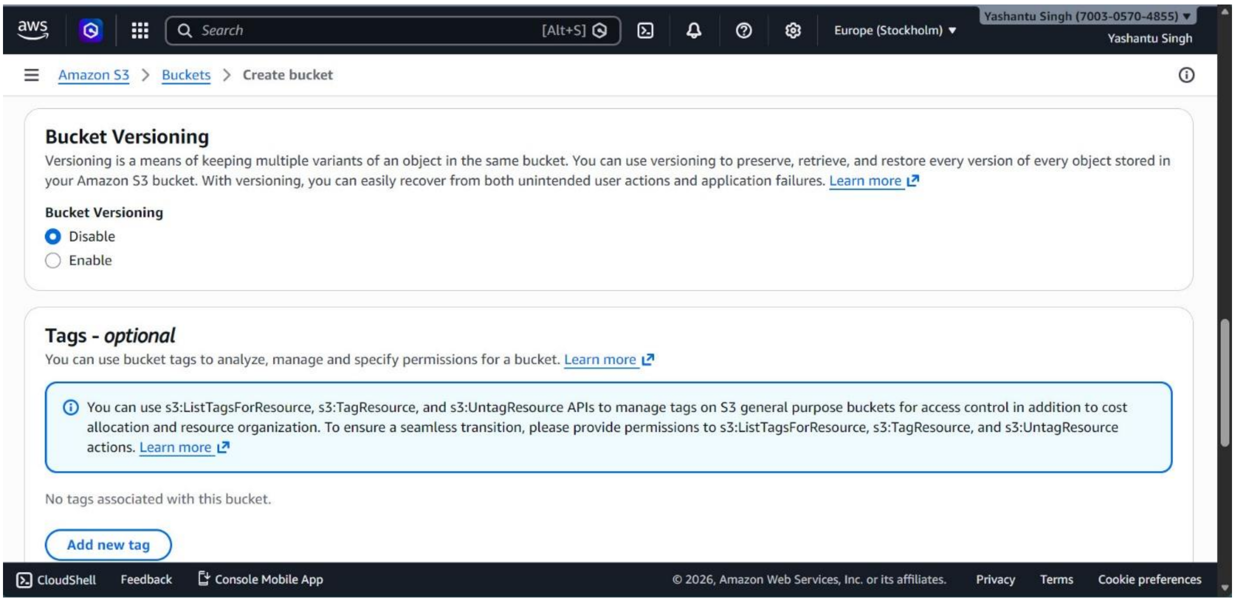
Step 6: Enable or disable bucket versioning

Bucket versioning allows you to preserve, retrieve, and restore every version of every object store in your bucket. It is recommended to enable versioning to protect against accidental deletions or overwrites.



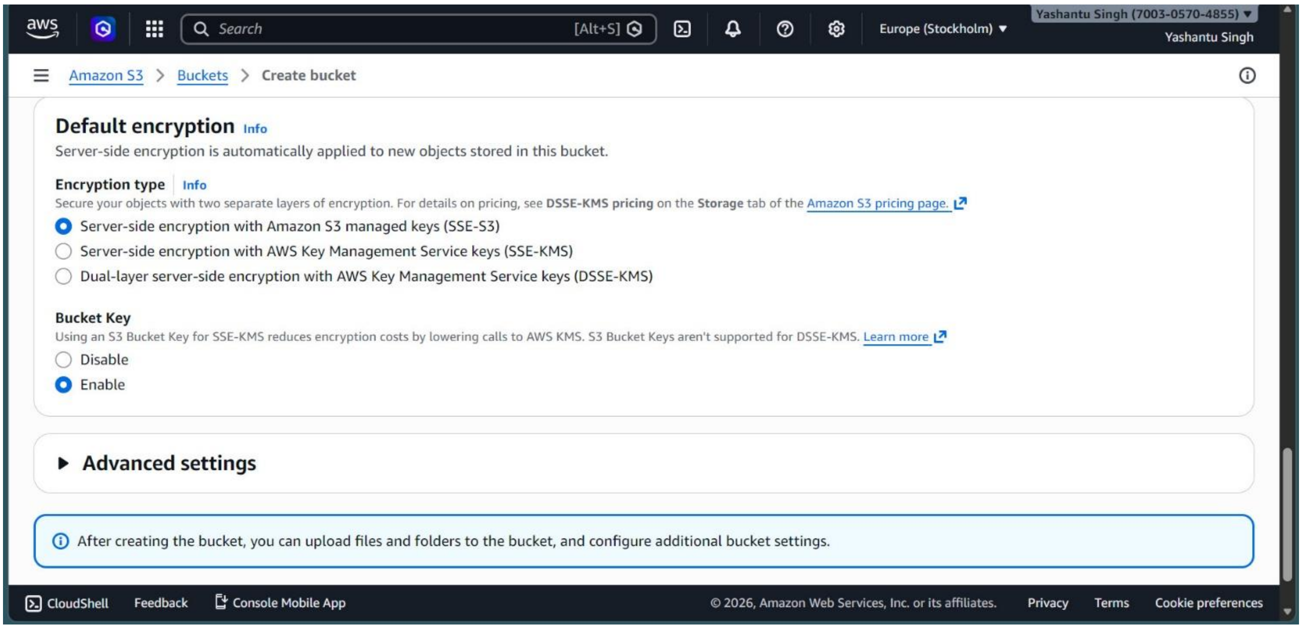
Step 7: Add tags for organization and cost allocation

Tags are key-value pairs that help you organize and manage your AWS resources. They can also be used for cost allocation and access control.



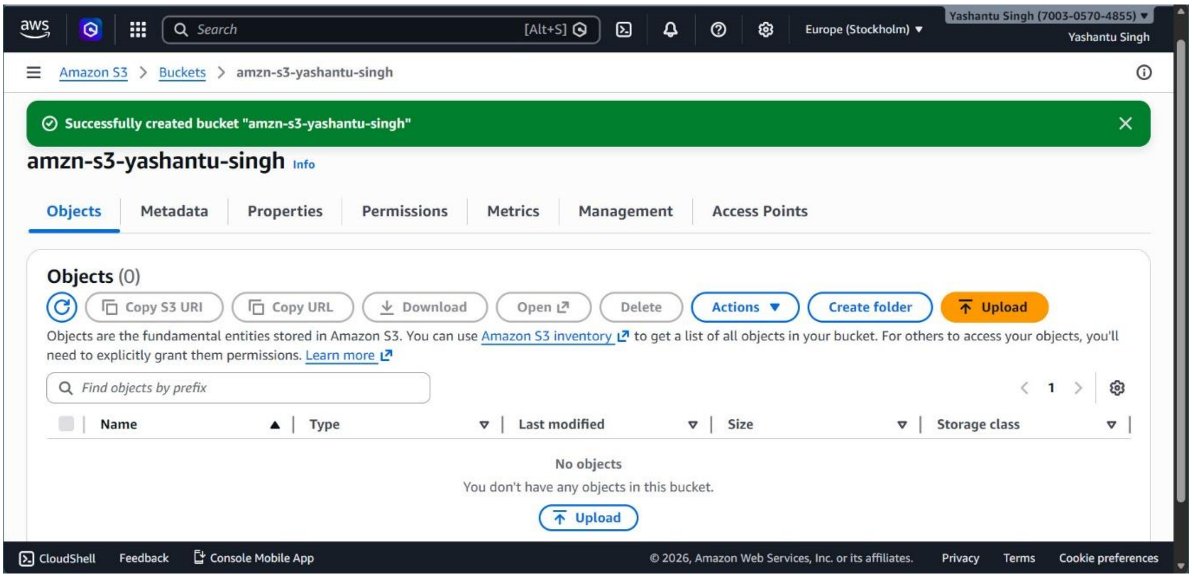
Step 8: Configure default encryption (SSE-S3, SSE-KMS, or DSSE-KMS)

Choose a server-side encryption method to protect your data at rest. Options include SSE-S3 (default), SSE-KMS (with AWS Key Management Service), or DSSE-KMS (dual-layer encryption).



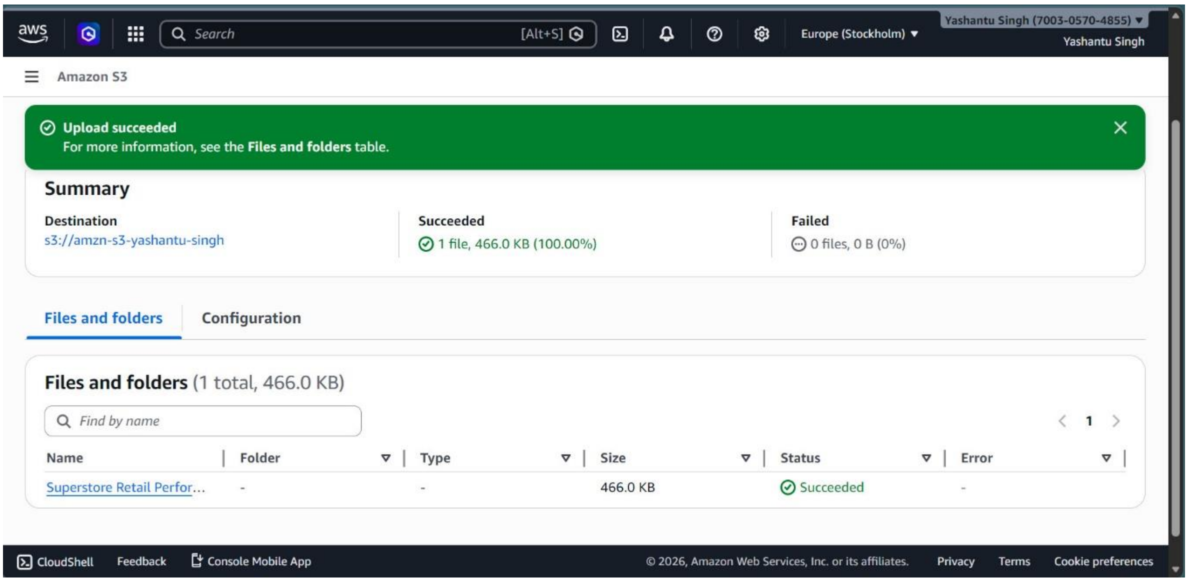
Step 9: Review and create the bucket

Review all your configurations. If everything looks correct, click the 'Create bucket' button to finalize the creation.



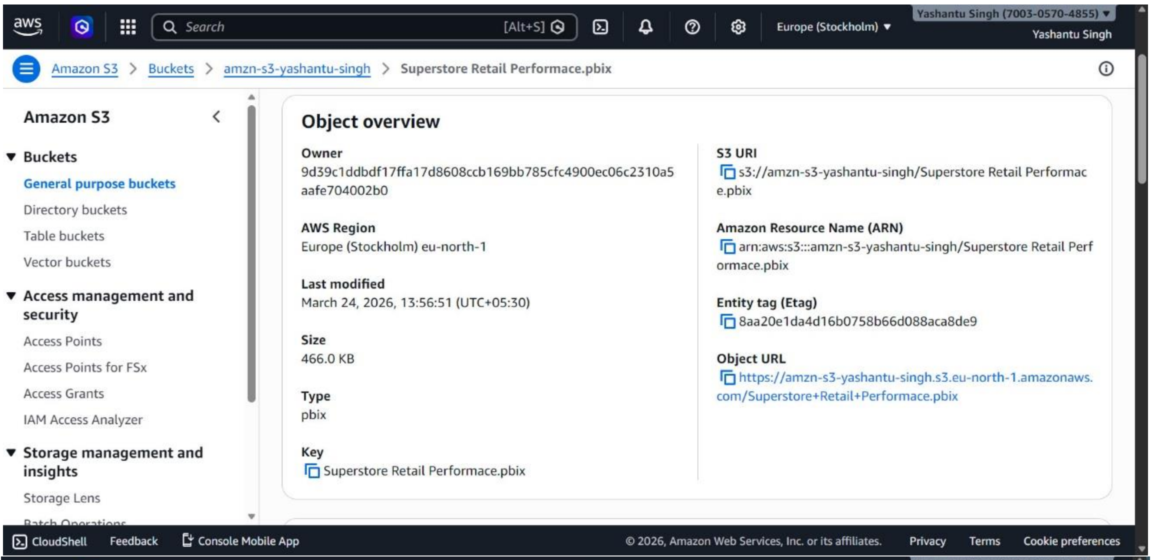
Step 10: Upload objects to the bucket and verify successful upload

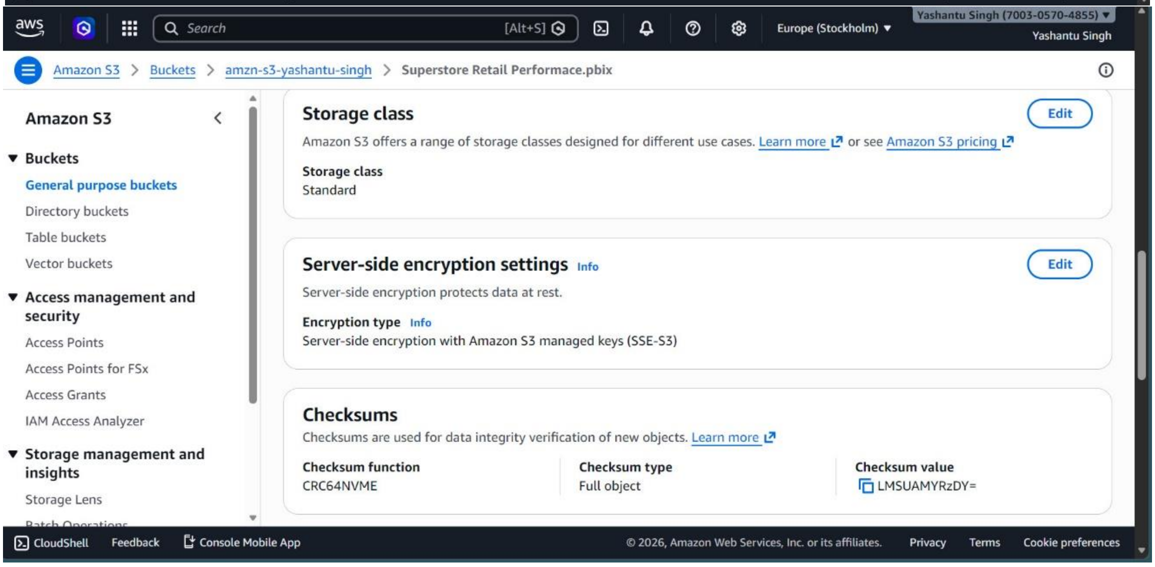
After creating the bucket, you can upload files using the 'Upload' button. Once uploaded, verify the success message and check the file list.



Step 11: View object properties (metadata, ARN, S3 URI, checksum, encryption)

Click on the uploaded object to view its properties, including metadata, Amazon Resource Name (ARN), S3 URI, encryption settings, and checksum for data integrity.





Step 12: Manage bucket lifecycle (versioning, replication, expiration rules)

Use lifecycle configurations to manage your objects. You can enable replication, set expiration rules, and manage versioning to optimize storage and data retention.